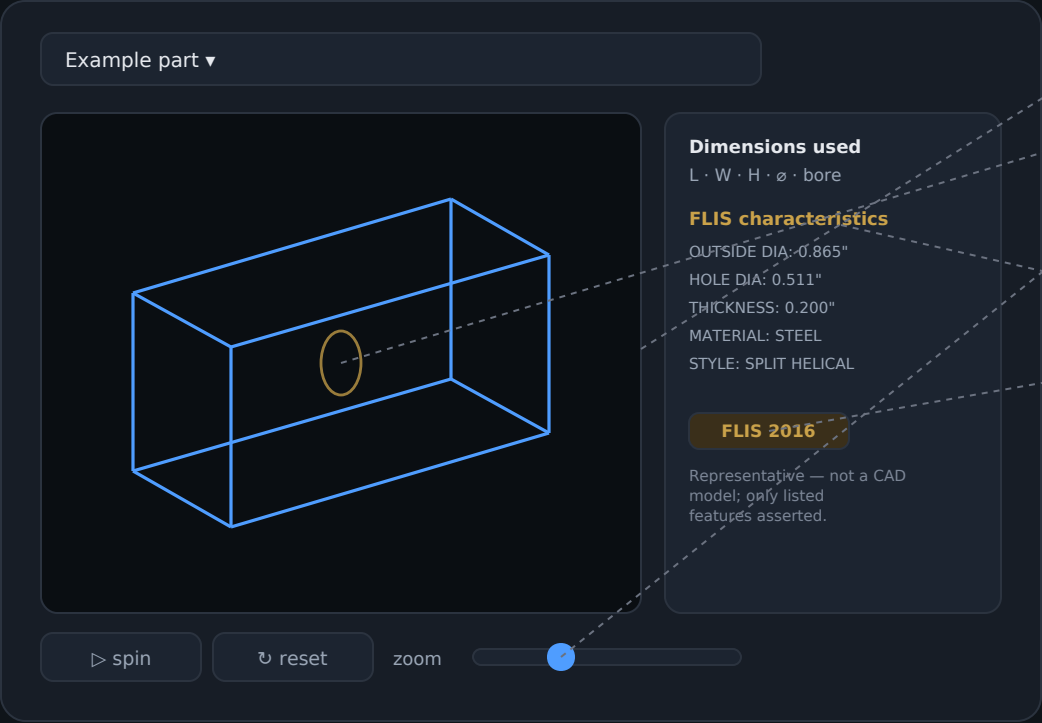


How the 2D→3D representative viewer works

It does not guess geometry. It scales a bounding solid to the dimensions FLIS states, lists the real features verbatim, and lets you rotate it.

THE VIEWER (marked up)



① bounding solid scaled to stated L×W×H

② amber ellipse = a stated bore (hole ø)

③ drag = rotate · scroll/slider = zoom · spin

④ the distinguishing features, verbatim & cited

⑤ FLIS vintage tag (year the data is effective)

HOW IT'S BUILT (per part)

1 · FLIS characteristics (cited)

'OUTSIDE DIAMETER: 0.865 IN; HOLE DIAMETER: 0.511 IN; THICKNESS: 0.200 IN...'

2 · Parse stated dimensions

regex → L, W, H, ø, bore (only what FLIS actually states)

3 · Bounding solid + bore

8 vertices of an L×W×H box + a centered ellipse for the hole

4 · Rotate + orthographic project

rotation matrix (drag sets angles) → 2D points

5 · Draw SVG wireframe

12 edges + bore → offline, no library, no WebGL

THE GROUNDING RULE (why this is honest)

- Model only what the documentation specifies. The solid is scaled to the dimensions FLIS states; nothing is invented.
- Surfaces are schematic — a bounding solid, not exact geometry. It conveys size + the stated bore, not unstated curves/features.
- The distinguishing features (hole count, thread, material, style) are shown VERBATIM next to it and cited to FLIS, with the data's year.
- So it's trustworthy exactly where it's labeled trustworthy — and a true CAD/photogrammetry model can drop in later for parts that need exact surfaces.